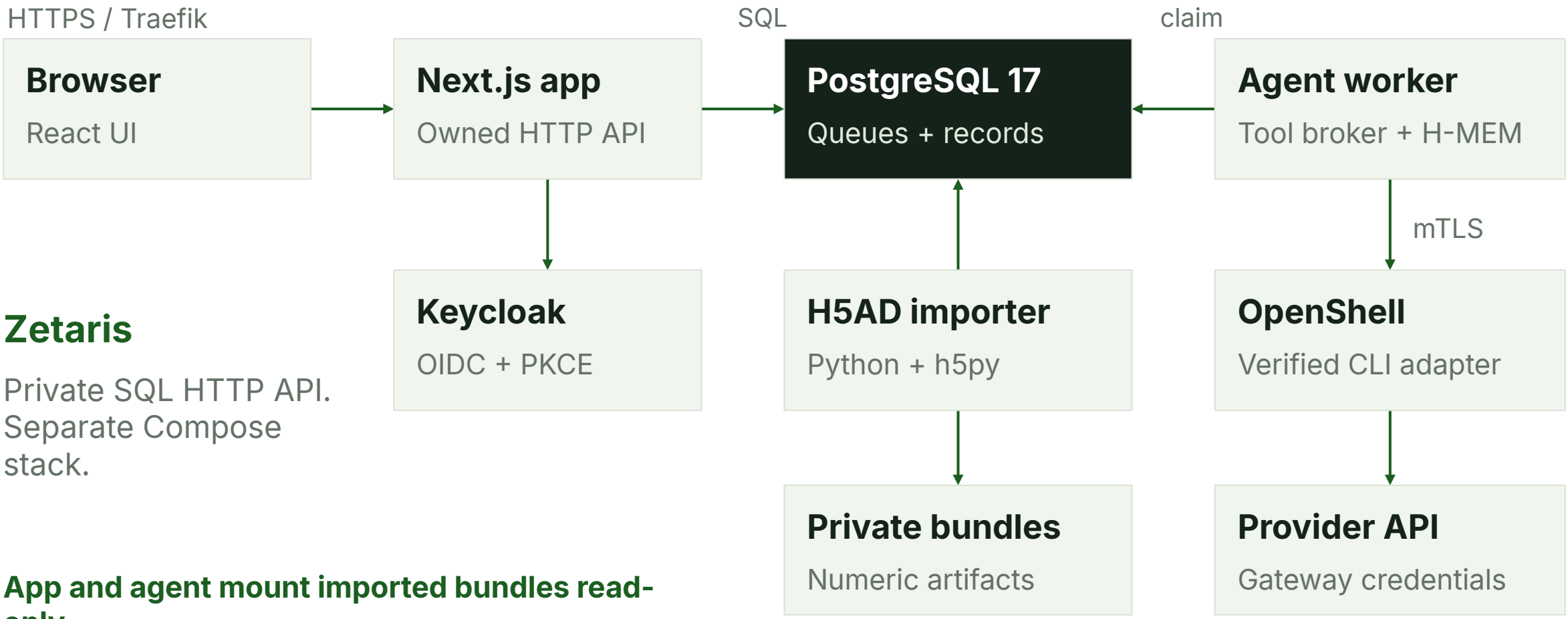
Deployment topology

Linux containers, PostgreSQL queues and a separate agent runtime boundary



Zetaris

Private SQL HTTP API. Separate Compose stack.

App and agent mount imported bundles read-only.

Agent runtime interface

The orchestration contract supports CLI execution and API-backed runtimes

Celloras orchestration

Owned context, tools, persistence, cancellation and card resolution

Adapter contract used by the worker

health()
run({ name, prompt, signal, progress })
Return a schema-valid AgentResponse

OpenShell

Codex CLI adapter
Verified end to end

Nightshift

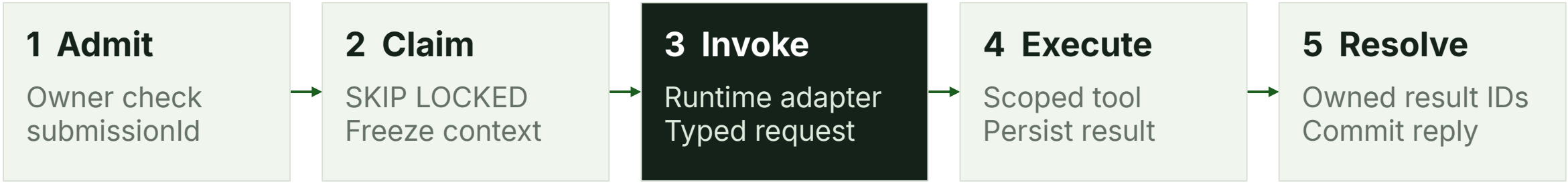
HTTP adapter implemented
Live acceptance pending

Extension targets

OpenAI / Anthropic APIs
OpenRouter, Vercel, other CLIs

Turn execution and typed output

PostgreSQL owns continuity across model invocations



After a tool result, invoke the runtime again with its saved result ID and summary.

AgentResponse v1

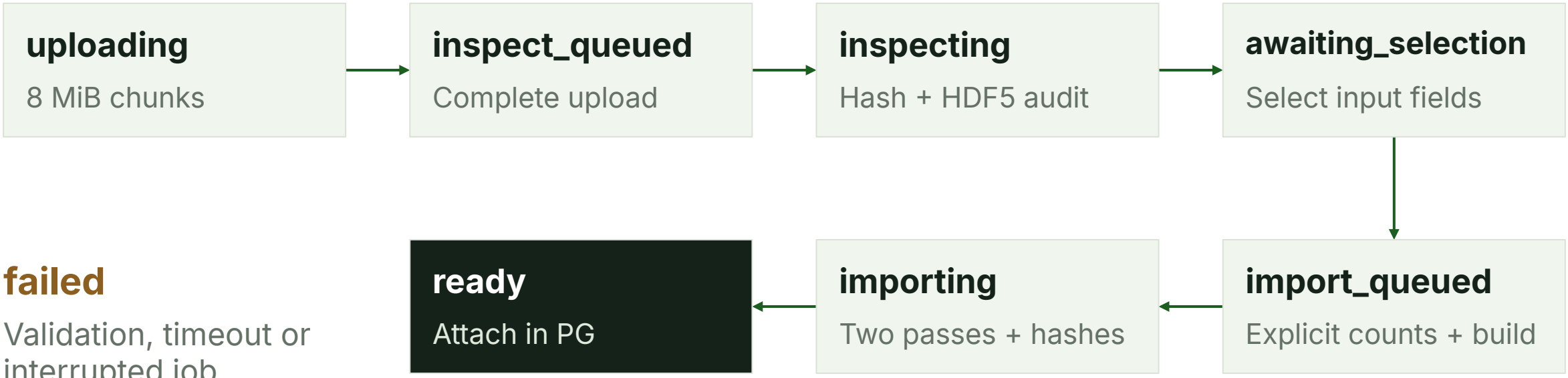
```
{ "version": 1, "blocks": [
  { "type": "markdown", "text": "..." },
  { "type": "annotation_result",
    "entryId": "<saved-run UUID>" }
]}
```

The server resolves canonical data

1–12 blocks. Strict keys and UUIDs.
At most 4 tools in a turn.
Cards use a fixed React registry.
Plot arrays stay in analysis_runs.

H5AD upload and import state machine

Resumable binary transfer, explicit field selection and bounded numerical conversion



PATCH headers

Upload-Offset + X-Chunk-SHA256
Sync file before committing offset.

Worker boundary

900 s / 3 GiB child process
Recheck file hash and owner before attachment.

Data model and ownership

An attached artifact version binds every scientific run to its input

PostgreSQL record	Durable contract
dataset_uploads	Offset, source SHA, selection, state and manifest SHA
project_datasets	Owned experiment + dataset ID + artifact version
agent_turns	Submission identity, frozen context and final response
analysis_runs	Operation, exact input version, result and provenance
annotation_reviews	Append-only decisions linked to an owned run + group

Immutable bundle on disk

Original H5AD + selection + manifest
Gene-major sparse counts, group aggregates and UMAP

Deterministic annotation pipeline

The agent selects and interprets tools. The tool computes candidate evidence.

1 Normalize per cell

$$y = \log1p(\text{count} / \text{cell total} \times 10{,}000)$$

2 Aggregate by existing group

Means include zeros. Detection is the fraction of cells with nonzero counts.

3 Score compatible reference genes

Population z-score across group means. Sum z-scores / reference gene count.

Support gate

Detection ≥ 10%
Group–rest delta ≥ 0.1
Gene z-score ≥ 0.5

Candidate gate

≥ 2 supporting genes
≥ 50% reference coverage
Mean score ≥ 0.5

Decision

Suggested, ambiguous or unresolved. Scores are exploratory statistics.

Zetaris reference protocol

A fixed SQL query supplies the versioned marker panel used by annotation

```
SELECT record_index, record_json, content_sha256
FROM CELLORAS.marker_reference_v1
WHERE reference_id = <pinned reference>
ORDER BY record_index
```

POST /query/sql/start

Read-only application account
pageLimit: 100
15 s timeout / 1 MiB limit

Validate

42 records in one page. Contiguous row indexes. Reconstructed document SHA.

Close

DELETE the query token in finally. Cleanup failure rejects the operation.

Persist

Request ID, retrievedAt, record count and SHA travel with analysis_runs.

Meterless H-MEM state and retrieval

The worker imports the pinned engine directly and persists its state in PostgreSQL



One transaction at turn claim

Record	Purpose
research_memory	Project engine state, sequence and pinned revision
research_memory_ledger	Append-only create / update / read events
agent_turns.context_snapshot	Exact recall supplied to that historical answer

Failure handling and observability

Every terminal response comes from an owned state transition

Failure	Implemented behavior
HTTP acknowledgement lost	Reuse submission ID and retrieve the existing turn
Worker dies or user cancels	Fail / cancel explicitly. No automatic inference replay
Sandbox cleanup fails	Retain lease and block the next turn until deletion
Upload bytes exceed PG offset	Truncate uncommitted tail before the next chunk
Reference or output validation fails	Reject the operation. Preserve canonical audit records

JSON activity logs

workerId, turnId, runId, event and status
Durable inputs, receipts and results remain in PostgreSQL.

Measured coverage and scaling limits

Public workflow receipt: 26 October 2026, application revision e153cf8

Boundary	Observed evidence	Remaining limit
H5AD ingestion	28.9 MB / 7,496 nuclei 40,127 genes / 12 groups	4 GiB cap configured Maximum-size load test pending
Numerical fidelity	Raw counts and UMAP exact 2,952 independent R checks	Independent biological validation pending
Sponsor path	42 Zetaris associations Saved review recalled via H-MEM	One reference table Explicit review memory only
Execution and storage	Real Linux agent workflow Ownership / resume checks	Single host + local volumes No multi-host throughput test

Next scale step: shared artifacts and coordinated worker scheduling.